中熔电气
SINOFUSE

ES05D-TMZ-N

产品规格书

PRODUCT SPECIFICATION

SINOFUSE

成为卓越的能量保护系统解决方案服务商
为整个人类的能量转换活动的安全运行提供保障
Provide excellent integrated solution of energy protection systems
Safeguard energy conversion processes for mankind

文件变更履历表/Revision Record

文件名称/File Name: ES05D-TMZ-N

文件编号/File No.: ZR/SCG-0015

Version	Content	Execution Date	Author
Y0	新制定/Initial release	2024-12-20	Wang Chengyu
Y1	增加630A规格及初版I-T曲线 Added 630A specification and initial version of I-T curve	2025-1-2	Wang Chengyu
S0	增加电流及对应电流参数与曲线 Increase current and corresponding current parameters and curves	2025-8-20	Wang Chengyu
S1	修正参数表、外形图及曲线字体 Correct parameter tables, outline drawings, and curve fonts	2025-9-15	Wang Chengyu

产品技术参数/Product Technical Parameters

●参考标准/Standards: IEC 60269-7
●额定电压/Rated voltage: DC 250V
●额定电流/Rated current: 200A~800A
●保护类别/Protection category: aBat
●分断能力/Breaking capacity: 50kA (时间常数/Time constant： 1～3ms)
●符合/Compliant: RoHS

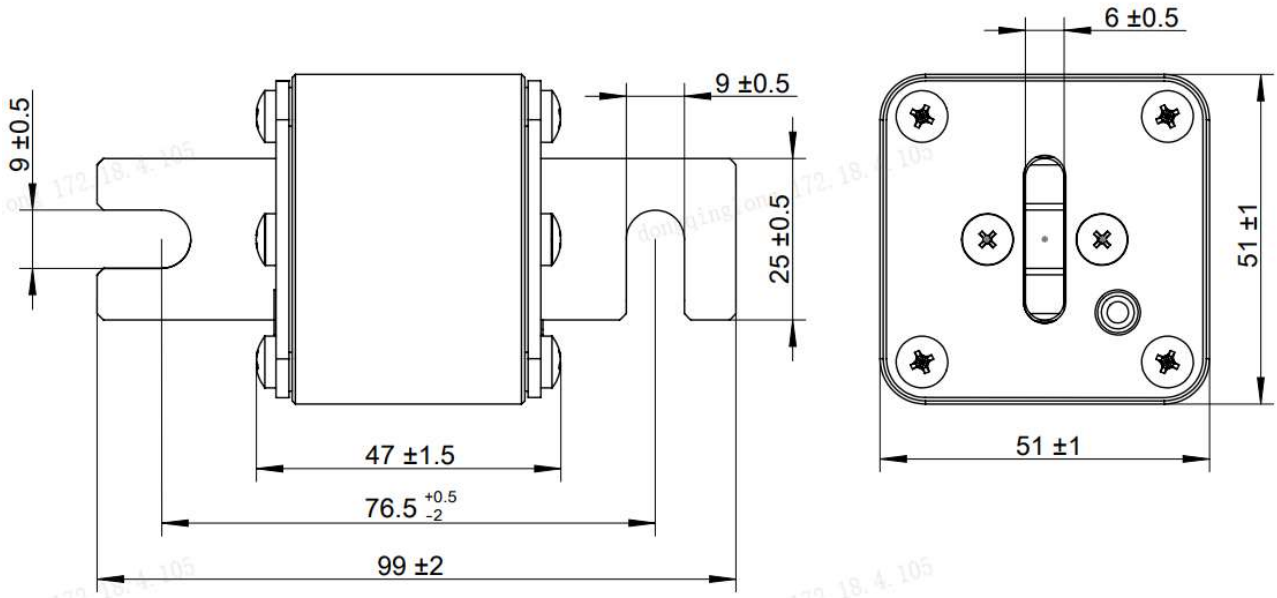
No.	Part Number	Size	In(A)	$I^2t/(A^2s)$		Loss(W)		Resistance Value mΩ (20℃)	Weight (g)	Max Package (pcs)	Remark
				Prearc	Clearing	0.5In	In				
1	ES05D-TMZ-N 200A	1	200	27900	50700	4.21	21.5	0.368±10%	400±5%	30	M8 11±1N·m
2	ES05D-TMZ-N 250A		250	43700	79300	5.29	27	0.294±10%			
3	ES05D-TMZ-N 315A		315	62900	114000	7.05	36.1	0.245±10%			
4	ES05D-TMZ-N 350A		350	85600	155000	7.49	38.4	0.210±10%			
5	ES05D-TMZ-N 400A		400	112000	203000	8.32	44.6	0.184±10%			
6	ES05D-TMZ-N 450A		450	185000	336000	8.48	45.2	0.142±10%			
7	ES05D-TMZ-N 500A		500	276000	501000	8.68	46.6	0.116±10%			
8	ES05D-TMZ-N 550A		550	328000	595000	9.55	49.3	0.106±10%			
9	ES05D-TMZ-N 630A		630	513000	931000	10.1	52.5	0.085±10%			
10	ES05D-TMZ-N 700A		700	699000	1270000	11.6	60.1	0.078±10%			
11	ES05D-TMZ-N 800A		800	1030000	1870000	12.5	65.3	0.064±10%			

注：表中I^2t数据均在电压DC257V标准试验环境下测得。
Note: All I^2t data in the table were measured under standard test conditions with a DC voltage of 257 V.

产品特性/Product Features

本产品为部分范围保护螺栓安装式熔断器，具有强限流、高分断能力等特点，可以作为储能系统、动力系统中半导体设备的短路保护。
This product is a partial-range protection bolt-mounted fuse with features such as strong current limiting and high breaking capacity. It can be used for short-circuit protection of semiconductor equipment in energy storage systems and power systems.
注：本规范中试验数据为标准环境下测得，试验参数和接线按照IEC 60269-7。
Note: The test data in this specification was obtained under standard environmental conditions, and the test parameters and wiring comply with IEC 60269-7.

产品外形尺寸/Dimensions(mm)

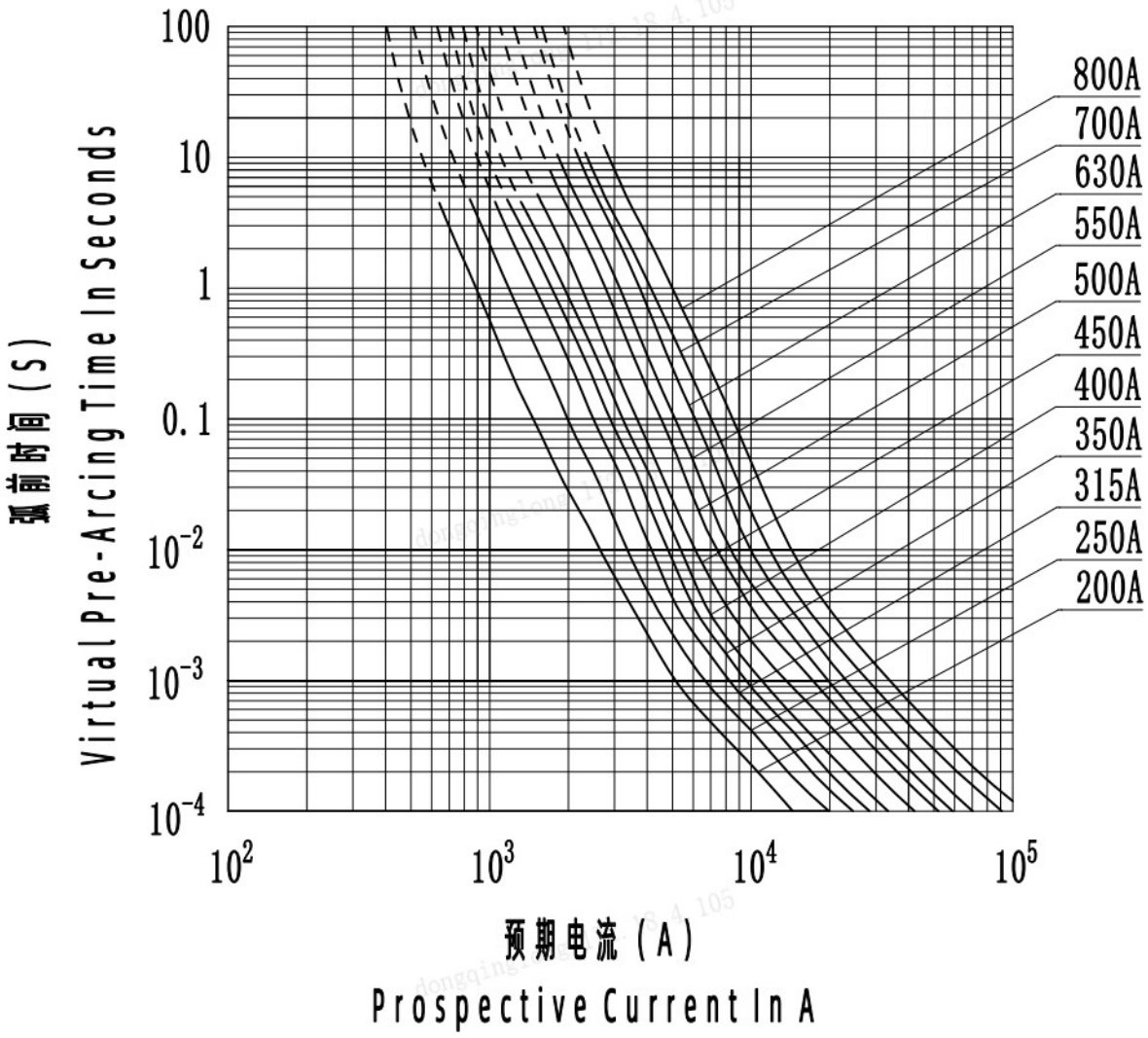


主要材质清单/Main Material List

No.	Part Name	Material	Electroplated
1	触刀/Tactile Knife	铜/Copper	镀暗锡/Tin-plated
2	瓷管/Porcelain Tube	陶瓷/Ceramics	/

产品特性曲线/Characteristic Curve

弧前I-t曲线/Pre-arcing I-t Curve



注/Notes：

1. 特性曲线精度为电流方向误差±15%；/Curves: current tolerance ±15%;
2. 最小分断电流≥3.5In；/Minimum breaking current ≥ 3.5In.
3. 曲线中100ms以下为等效弧前时间。/Below 100ms in the curve is the equivalent pre-arc time.

运输、存储/Transportation, Storage

运输/Transportation

避免过度挤压或冲击：在运输过程中，需要避免对熔断器施加过度挤压或直接冲击，这可能会损坏熔断器的外壳、接线端子或内部元件，从而影响其性能和安全性。运输过程中应避免雨雪侵袭和机械损伤。

Avoid excessive compression or shock: during transportation, it is necessary to avoid applying excessive compression or direct shock to the fuse. this may damage the fuse housing, terminals or internal components, thus affecting their performance and safety. avoid rain, snow and mechanical damage during transportation.

防潮防湿：熔断器应远离潮湿和湿润的环境，以防止受潮和腐蚀。必要时，可以使用防潮包装材料或采取其他措施来保护熔断器免受潮湿影响。

Moisture and humidity protection: fuses should be kept away from damp and wet environments to prevent moisture and corrosion. if necessary, moisture-proof packing materials may be used or other measures taken to protect the fuse from moisture.

存储/Storage

1.产品存储温度：-40℃～85℃；
Storage temperature: -40℃～85℃.
2.包装存储温度：-40℃～70℃；
Package temperature: -40℃～70℃.
3.包装存储湿度：相对湿度≤90%，无凝露。
Package humidity: RH≤90%, no dewing.

使用条件/Use Conditions

熔断器所有试验均在标准测试条件下进行，因实际使用条件和标准测试条件存在差异，在选用时需考虑其影响。当实际使用条件满足允许使用条件时，需按照对应修正曲线对熔断器的额定值进行修正；当实际使用条件不满足允许使用条件时，请咨询我司。

All tests for the fuse are conducted under standard testing conditions, there is the difference between the actual use conditions and the test conditions, the influence of which should be taken into account when selecting the fuses. when actual usage condition meets tolerable range, the rated value should be corrected according to the corresponding correction curve. otherwise, please consult Sinofuse.

大气条件/Atmosphere

空气是干净的，相对湿度在最高温度为40℃时不超过50%。在较低的温度下可以有较高的相对湿度，例如，在20℃时，相对湿度可达90%。
For clean air, maximum 50% RH at 40℃. higher RH is allowed when temperature is lower, e.g. maximum 90% at 20℃.

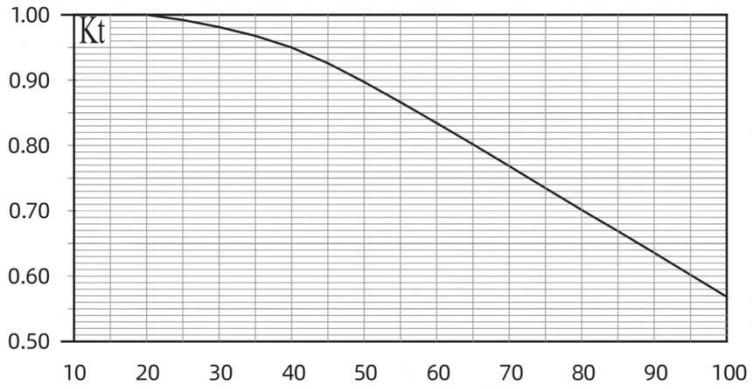
周围空气温度/Ambient Air Temperature

标准测试条件：20℃；
Standard testing: 20℃.
允许使用条件：-40℃～100℃；
Tolerable range: -40℃～100℃.

熔断器在周围空气温度20℃以上工作，额定电流按照下面曲线进行修正；温度修正时一般不考虑增加熔断器额定电流（除非20℃以上不是工作范围）。

When the fuse operates at an ambient air temperature above 20℃, the rated current shall be adjusted according to the following curve. temperature derating typically does not consider increasing the fuse's rated current (unless above 20℃ is not within the operating range) .

周围空气温度修正曲线/Correction Curve of Ambient Air Temperature



海拔/Altitude

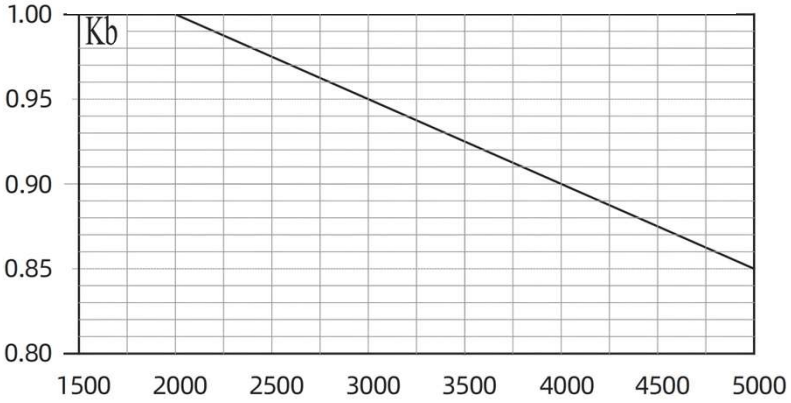
标准测试条件：海拔≤2000m；

Standard testing: altitude≤2000m.

允许使用条件：海拔≤5000m，不同海拔下熔断器额定电流按照下面曲线进行修正。

Tolerable range: altitude≤5000m; rated current should be corrected according to the following curve at different altitudes.

海拔修正曲线/Altitude Correction Curve



风冷/Air Cooling

标准测试条件：无风，周围1m以内无发热器件；

Standard testing: no ventilation, no heat source within 1m.

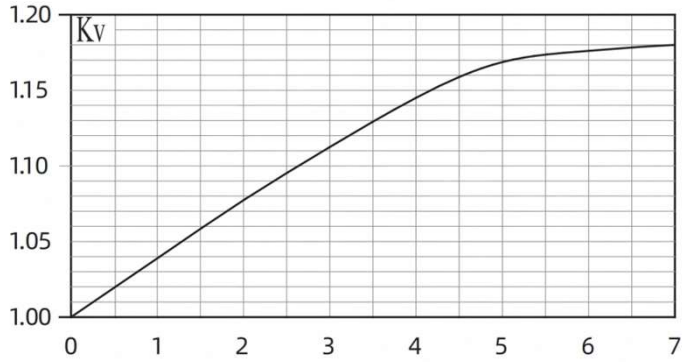
允许使用条件：允许有风及周围有发热器件，但需考虑其影响；

Tolerable range: ventilation and heat source allowed, but their impacts should be considered.

强制风冷：采用强制风冷可以改善熔断器的散热，提升熔断器的额定电流，强制风冷条件下熔断器额定电流按照下面曲线进行修正。

Forced air cooling: forced air cooling improves the heat dissipation of fuses, enabling an increase in their rated current. Under forced air cooling conditions. rated current should be corrected according to the following curve.

强制冷却风速修正曲线/Correction Curve of Forced Cooling Air Velocity



强制冷却风速/Forced Cooling Air Velocity (m/s)

铜导线/铜排尺寸/Copper Wire/Busbar Size

标准测试条件：铜导线/铜排尺寸满足标准要求；

Standard testing: the size of the copper wire/busbar complies with the standard requirements.

允许使用条件：推荐铜导体截面不小于标准截面的50%（标准截面以实际通流值为前提），其他材质导体需换算为铜导体的尺寸进行校核。

Tolerable range: it is recommended that the cross-sectional area of the copper conductor should not be less than 50% of the standard cross-sectional area (the standard cross-section is based on actual current-carrying capacity). for conductors made of other materials, their dimensions must be converted to equivalent copper conductor dimensions for verification.

标准导线选用表（参考GB/T13539.1）/Standard Conductor Selection Chart（GB/T13539.1）

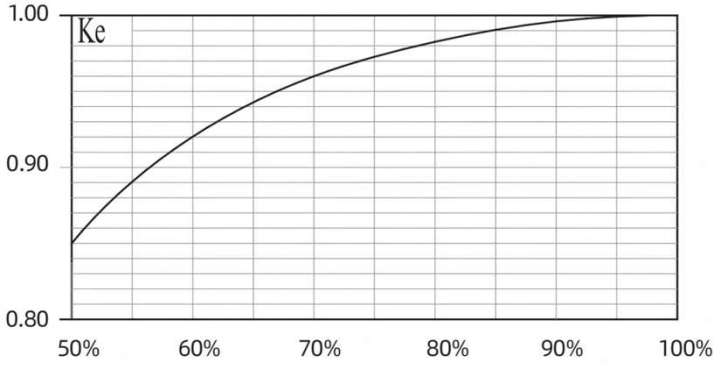
Rated Current A	Conductor Cross-sectional Area mm^2
2、4、6	1
8、10、12	1.5
16、20	2.5
25	4
32	6
40、50	10
63	16
80	25
100	35
125	50
160	70

Rated Current A	Conductor Cross-sectional Area mm^2
200、225	95
250、300	120
315、350	185
400、425	240
500	300
630	370
800	480
1000	600
1250	800
> 1250	(1.0~1.6)A/mm^2

不同铜导体尺寸下熔断器的额定电流按照下面曲线进行修正

Rated Current Should Be Corrected According to The Following Curve

接线修正曲线/Wiring Correction Curve



连接导线截面/标准截面

Connecting Conductor Cross-section/Standard Cross-section

机械振动/机械冲击/Mechanical Vibration and Shock

非底座安装类熔断器有良好的抗振动和冲击的耐受能力（满足GB/T 21563 轨道交通机车车辆设备 冲击和振动试验1类B级）。

Non base-mounted type fuse has great resistance to vibration and shock (meets Standard GB/T 21563 Shock and Vibration Test Ⅰ.B Level).

关于底座安装类安装熔断器的机械振动/冲击能力，以及方体瓷管类产品机械振动/机械冲击更为强烈的应用环境，请咨询我司。

For the mechanical vibration/shock capabilities of base-mounted fuses and harsher conditions for square porcelain tube products, please consult Sinofuse.

污染等级/Pollution Level

熔断器满足污染等级2（详见GB 14048.1-2023 7.1.3.2）。

Fuse meets pollution level Ⅱ (see GB 14048.1-2023 7.1.3.2 for details).

熔断器安装条件/Mounting Condition

使用规格书推荐扭矩安装，安装方式及注意事项见相应结构熔断器安装说明。

Use the torque specified in the specification. refer to the fuse installation guide for methods and precautions.